

Eventless Payload-Adaptive NMPC for Quadrotors

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Abstract—A release command does not guarantee physical payload detachment, so switching a controller’s model from an attach/drop signal can be unsafe. We present an eventless payload-adaptive control framework in which a moving horizon estimator (MHE) [1] uses only motor speeds and odometry to publish an atomic, health-gated frame of composite mass, first mass moment $s = m_{prxy}$, and inertia to an outer-loop nonlinear model predictive controller (NMPC). The first-moment representation remains well defined as the payload vanishes, while bounded filtering and continuous body-rate scheduling prevent discontinuous model updates. A thrust-residual self-trigger deweights stale pre-change stages in the MHE window and reduces mass-estimate settling time by up to $4\times$. In PX4 [2] software-in-the-loop experiments with physically attached eccentric payloads, the eventless interface completes release confirmation in 21/21 valid paired flights, with no observed post-release instability, at a median confirmation delay of 3.76 s when the moment vote clears in maneuver and 16.46 s when a brake to hover is first required. A safety ablation shows that payload quantity and thrust residual are individually unreliable during aggressive flight: their fast release branch causes premature model clearing in 3/9 flights. Requiring fresh moment-collapse evidence eliminates premature declarations in a 12-flight smoke test, yielding 11 confirmations and one conservative unresolved outcome. Against an L1-adaptive [3] NMPC baseline, the proposed method provides comparable transient tracking, recovers 15% to 35% faster, and retains interpretable physical parameters. These results demonstrate the eventless control interface and its fail-safe semantics, while not claiming distribution-wide robustness of the release detector.

I. INTRODUCTION

Autonomous aerial delivery and in-flight aerial manipulation subject a multirotor to abrupt changes in its own dynamics. Grasping or releasing a parcel changes three coupled quantities at once: the total *mass* m (which rescales the thrust-to-acceleration map and the hover throttle), the *center of mass* (a thrust that no longer passes through the composite CoM produces a constant trim torque $\tau \approx \Delta r \times f$, significant even for small offsets because $f \approx mg$ is large), and the *rotational inertia* (which slows the attitude response and effectively lowers the inner-loop bandwidth). The first channel is an outer-loop (position/thrust) problem; the latter two are inner-loop (attitude/rate) problems. A controller that models the payload only as a scalar mass will mistrack or, for sufficiently heavy eccentric loads, destabilize.

Three families of methods address load changes. *Explicit parameter estimation* (EKF/MHE of mass, CoM, inertia) is interpretable, integrates natively with a model-based controller, and can update thrust-margin constraints, but classical estimators converge slowly across an abrupt change because the estimation window or filter averages across the discontinuity. *Lumped-disturbance adaptation* (L1-adaptive NMPC, INDI)

compensates the aggregate effect without estimating physical parameters, reacting fast but yielding no interpretable model. *Learning-based* methods (e.g. neural MHE that regresses estimator weights) can resolve the estimator’s bias–variance trade-off but typically require rich sensing or offline ground-truth data.

The central question of this work is whether the complete grasp–carry–release lifecycle can adapt its payload model without trusting a commanded or externally supplied payload event. Our answer rests on two observations. First, an abrupt payload change leaves a detectable common-mode signature in reconstructed collective thrust, so the estimator can discover *when* the dynamics changed without consuming a gripper command, attachment state, or external event notification. Second, **motor speeds are a free physical sensor**: both collective thrust and body torques can be reconstructed from rotor speeds alone, yielding physical residuals that are decoupled from the estimator’s own state. Combining the two yields an estimator that is both fast (it deweights stale pre-event measurements the moment the thrust residual crosses a threshold) and *sensing-minimal* (it needs no external event signal and no measurement of the attachment geometry).

Contributions.

- 1) An **eventless payload-adaptive NMPC architecture** whose model never consumes an attach/drop notification or treats its own release command as evidence. An atomic, health-gated $m/s/J$ frame drives bounded model updates and continuous body-rate scheduling (§III–IV).
- 2) A **self-triggered MHE** that detects abrupt payload changes from reconstructed thrust and deweights stale pre-event window stages, reducing mass-estimate settling time by up to $\sim 4\times$ without an external event signal. The first-moment state $s = m_{prxy}$ represents eccentricity without measured attachment geometry and remains well-defined at release (§IV; §VI-A).
- 3) **Closed-loop lifecycle and safety validation**: 21/21 valid paired flights complete eventless release without observed post-release instability. A targeted ablation rejects an unsafe quantity-plus-residual branch and shows that fresh moment-collapse evidence with an UNRESOLVED fail-safe state is necessary (§VI-C).

The runtime estimator consumes only motor speeds and odometry; the sole geometric quantity not identified online is a weak vertical-arm prior r_z , which is structurally unobservable from the thrust-moment channel.

II. RELATED WORK

Agile quadrotor flight. Modern quadrotor control builds on geometric tracking on SE(3) [4], and pushing to the actuator limits is by now standard in time-optimal flight [5]. Both assume the inertial parameters are known and fixed; a payload event invalidates that assumption mid-flight, which is the gap this work addresses.

Aerial manipulation and transport. Aerial robots that physically interact with their environment—grasping, carrying, and releasing objects—are surveyed in [6]; dynamic aerial grasping, where the gripper dynamics must be explicitly modeled and controlled, was demonstrated in [7]. Load transportation specifically—by grasping or by cable suspension—is surveyed in [8]. Our setting is the transport-and-release subproblem, where the payload is rigidly attached and the resulting change in mass, inertia, and center of mass must be handled by the controller.

Control under changing payload. Adaptive and robust designs—L1-adaptive NMPC [9], INDI [10]—treat the load as a lumped disturbance and are the standard baselines for payload-carrying flight; L1-NMPC in particular reports large tracking-error reductions under unknown load on aggressive trajectories. A parallel line learns the residual dynamics instead: [11] fits unmodeled aerodynamic effects with Gaussian processes inside the MPC, which improves tracking without yielding a named physical parameter either. Slung-load control addresses the swinging-payload case explicitly [12], exploiting differential flatness of the cable-suspended system; a rigidly attached payload instead changes the vehicle’s own inertial parameters. A related proprioceptive line estimates the external wrench directly with momentum-based observers, enabling collision detection and interaction control [13]. All of these recover a *force/torque*, not a physical parameter: they cannot, e.g., update a thrust-margin constraint or an inner-loop gain schedule from the estimate.

MHE and online parameter estimation. Filter- and optimization-based estimators recover mass, inertia, and geometric parameters [14]–[16]. Standard MHE performance is highly sensitive to the process/measurement weight matrices; neural MHE (NeuroMHE) [17] learns to output these weights online, resolving the bias–variance trade-off but estimating a lumped disturbance force rather than explicit named parameters.

Event-triggered and change detection. Event- and self-triggered control schedule computation by state-dependent conditions rather than by a fixed clock [18]; our trigger is of the self-triggered kind in that the condition is evaluated from signals the estimator already has. Residual-based fault/change detection (CUSUM and related [19]) underlies it. Our positioning: we do not learn dynamics and add no sensing; every deployed signal derives from motor speeds and odometry. This is a deliberate departure from learning-based MPC [20] and the broader safe-learning agenda [21], where a learned component absorbs the model mismatch; here the mismatch is resolved into named parameters that downstream constraints

can consume. We separately test, as a supporting ablation, whether numerical tuning improves the handcrafted transient-period schedule.

III. SYSTEM OVERVIEW AND PROBLEM SETUP

Architecture (Fig. 1). Three layers: Gazebo physics (with a magnetic DetachableJoint gripper producing genuine inertia/CoM changes), the PX4 inner-loop rate controller, and a ROS 2 outer loop (NMPC + MHE). The outer loop publishes a collective thrust and a body-rate setpoint; PX4 tracks the body rate.

Dynamics and parameterization. The NMPC model carries the runtime parameter vector $p = [x_{\text{ref}}, m, \Delta J, c_{xy}]$ (extended with a lumped disturbance only for the L1 baseline). Thrust acting at the geometric center about the composite CoM produces $\tau_{\text{com}} = [-c_y T, +c_x T, 0]$. The deployed MHE augments both composite mass and the horizontal first mass moment,

$$x_{\text{MHE}} = [x; m; s_x; s_y] \in \mathbb{R}^{16}, \quad s = mPr_{xy}, \quad (1)$$

with $\dot{m} = 0$ and $\dot{s} = 0$ within each window ($N = 20$, $\Delta t = 0.1$ s). The published geometry is reconstructed as

$$c_{xy} = \frac{s}{m}, \quad \mu = \frac{m_B \max(m - m_B, 0)}{m}, \quad (2)$$

$$\Delta J_{xx} = \Delta J_{yy} = \mu r_z^2.$$

where r_z is a weak rack prior. The small $O(r_{xy}^2)$ diagonal terms and yaw increment are omitted in the deployed interface. The hover-input baseline is recomputed from the live model and includes the eccentric trim torque, avoiding a stale operating-point bias.

Model-independent physical inputs. We reconstruct the collective thrust $T_{\text{phys}} = k \sum_i \omega_i^2$ and the body torques from rotor speeds and arm geometry. The quadratic coefficient k is a static-thrust constant; it is known to lose validity under large induced inflow during fast translation [22], which we do not model and which is one source of the estimator bias discussed in Sec. VII. Feeding the NMPC’s *commanded* input back into the estimator instead creates a self-consistent blind spot: at hover $m \approx T/g$ is self-consistent for any m , so the mass is unobservable from the commanded thrust. The motor-speed reconstruction breaks this because it is decoupled from the controller/estimator state.

Thrust-to-throttle inversion. The offboard collective-thrust command is inverted to a normalized rotor command through the motor model $T = k\omega^2$; because k appears on both the forward and inverse paths, steady-state tracking is independent of the absolute thrust calibration.

Continuous inner-loop scheduling. Model consistency in the NMPC cannot by itself restore the PX4 rate loop’s lost bandwidth. The deployed path therefore maps the live inertia increment to a body-rate-command scale, low-pass filters it, applies target hysteresis and asymmetric rise/recovery slew limits, and preserves 5% command headroom before the final saturation. It does *not* rewrite PX4 parameters. An attach-time inertia-only bootstrap uses the rack’s 0.30 kg payload envelope

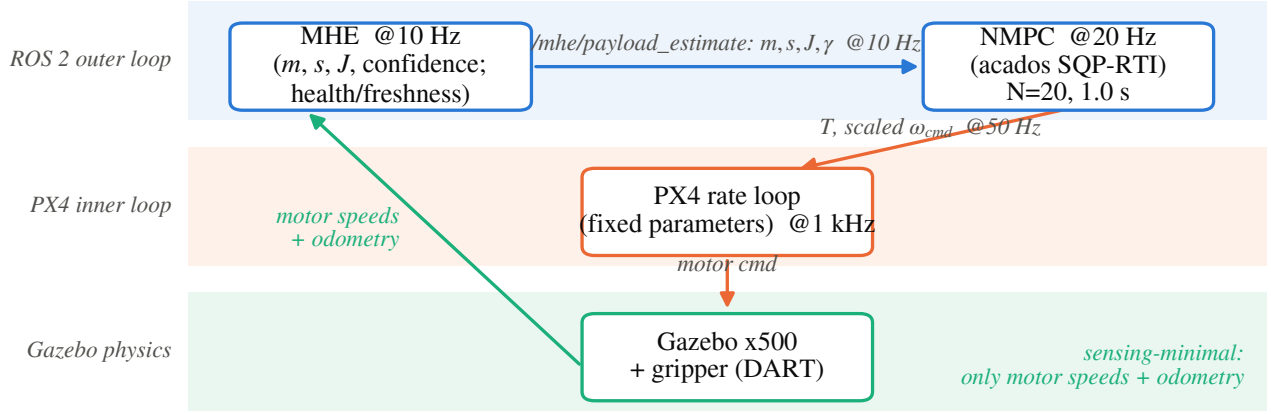


Fig. 1. System architecture and information flow. The full pipeline consumes only motor speeds and odometry; one atomic, health-gated $m/s/J$ frame drives both the NMPC model and continuous rate-command scheduling. No attach/drop notification enters this path.

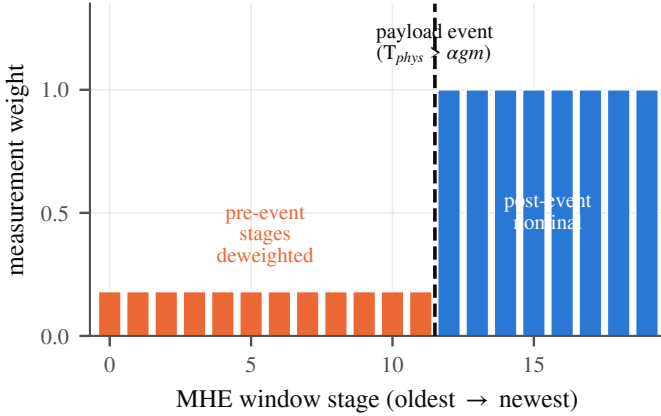


Fig. 2. Sliding-window deweighting. On a payload event (thrust residual crossing αgm), the pre-event window stages are deweighted so stale cross-discontinuity measurements no longer pull the mass estimate.

for immediate safety, then fades out after persistent coherent MHE evidence; neither payload mass nor s is injected by this bootstrap, and the drop path remains estimate-only.

IV. SELF-TRIGGERED ESTIMATION AND EVENTLESS ADAPTATION

A. Deweighting mechanism

Within the sliding window, upon a payload event we deweight the R/Q blocks of the pre-event stages (per-stage `cost_set`), rather than softening the mass anchor Q_0 (which is already soft and is not the bottleneck; the true bottleneck is the $\dot{m} = 0$ rigidity that lets stale cross-discontinuity measurements pull the mass). Guardrails: the terminal measurement never enters the cost, and the first solve after an event keeps nominal weights.

B. No-signal self-trigger

The detector arms only after the baseline has settled (a warmup gate of consecutive quiet thrust-residual frames),

then fires when $|T_{\text{phys}} - \text{baseline}|$ exceeds a threshold for a persistence count of consecutive frames. The warmup gate rejects the takeoff transient; the persistence count rejects gust pulses (broadband/transient) while catching a sustained mass step. This same criterion is the false-trigger criterion characterized under wind in §VI. This detector schedules the MHE window weights. Release confirmation uses the multi-source logic below; the two roles must not be conflated, because a threshold suitable for a 0.30 kg schedule event can suppress the residual vote of a 0.15 kg release.

C. Continuous payload interface and autonomous lifecycle

The MHE publishes one atomic frame

$$\mathcal{P} = \{m, s_x, s_y, c_x, c_y, \Delta J_{xx}, \Delta J_{yy}, \Delta J_{zz}, J_{xx}, J_{yy}, J_{zz}, \gamma, h, a\}, \quad (3)$$

where γ is no-payload confidence, h is a health bit and a is solution age. Health requires a successful latest solve and fresh odometry, motor-speed, and known-input streams. The NMPC additionally checks local receive age; an unhealthy or stale frame freezes the last accepted model and breaks the confidence persistence timer. Valid frames pass through a low-pass filter and separate slew limits for m , s , and ΔJ before every NMPC solve.

Payload presence is an estimator-owned state, deliberately distinct from the legacy flag that means “an external attach notification was received.” A hysteretic `EMPTY` ↔ `LOADED` machine uses sustained payload-mass evidence, with a directional residual as a fast path, and a separate latch records that the present flight reliably entered `LOADED`. This latch prevents an empty-flight confidence value from consuming the later drop confirmation and prevents a transient mass dip during a figure-eight from disarming the true release path.

No-payload confidence is the product of smooth empty scores for payload quantity, first moment, and inertia. A nonzero moment therefore vetoes a false empty decision even if the mass estimate touches its lower bound. On accepted

release evidence, the estimator switches only the *published target* of s to zero; the optimization state is not overwritten. The output filter consequently produces a continuous decay rather than a geometric step. The controller marks a planned drop complete, resets its L1 state, and finishes recovery only after healthy confidence remains above 0.90 for 1.0s; issuing the gripper command itself does none of these things.

D. Dimensionless trigger schedule

The deployed hand-designed schedule M0 applies a fixed pre-event deweighting rhythm and expresses the confirmation threshold as the dimensionless ratio α , with physical threshold αgm . This preserves the same relative margin across payload masses, whereas a fixed threshold in newtons does not. A supporting simulation-in-the-loop ablation tested whether optimizing the schedule magnitude improves M0; Sec. VI-B summarizes the negative result. M0 is retained in all reported system-level experiments.

E. Why the first mass moment is the essential state

Estimating CoM offset alone is ill-conditioned at release: physically $s = m_P r_{xy} \rightarrow 0$, whereas r_{xy} itself is undefined as $m_P \rightarrow 0$. In an earlier mass-only formulation, the optimizer could suppress a phantom eccentricity only by driving total mass below the bare-airframe value; 14 of 55 post-release samples then pinned at the lower bound. Augmenting s supplies the rotational channel directly and permits $c_{xy} = s/m$ without knowing the grasp-dependent horizontal offset. A frozen-window inertia coefficient prevents the optimizer from using instantaneous mass as a free inertia knob, while the weak r_z prior retains the dominant parallel-axis term. This structure is the minimum one that simultaneously represents payload existence, trim moment, and roll/pitch inertia without an external attachment geometry measurement.

The release detector studies three physically distinct sources: (i) collapse of $\rho = \|s\|/\|s\|_{\text{peak}}$, with an absolute $\|s\|$ bound used only as a sanity check; (ii) payload quantity, in which mass and algebraically derived inertia count as *one* source; and (iii) a directional thrust-residual edge with finite validity. A single source never releases. The slow path requires quantity-empty and moment-collapse evidence for five MHE frames. The release-only residual is deliberately separate from the 4.4N schedule-confirmation threshold and uses a signed two-window difference, $\Delta T = \bar{T}_{\text{post}} - \bar{T}_{\text{pre}}$, with 0.3s half-windows. We evaluated two residual branches: fastA paired a normal 0.9N residual with quantity-empty evidence, while fastB paired a longer-persistent residual with moment collapse. A recorded figure-eight transient reached -1.69N , larger in magnitude than the -1.47N step of the minimum supported payload, so amplitude alone cannot separate the cases. Online and offline evaluation call the same pure decision function. The randomized smoke of Sec. VI-C rejects fastA; the safety-accepted logic therefore requires moment-collapse evidence (slow or fastB) and treats an unresolved timeout as unknown, never as proof of an empty vehicle. Moment evidence must

also originate from a fresh, successful MHE solve: the re-anchoring routine deliberately initializes s to zero after five consecutive solver failures, so that initialization is invalid evidence rather than a physical collapse.

V. SIMULATION SETUP

PX4 SITL with Gazebo [23] (Harmonic), an x500 quadrotor, and a magnetic DetachableJoint gripper that produces genuine inertia and CoM changes.

Two payload-change mechanisms, by design. The physical DetachableJoint gripper attaches a separate rigid body and therefore couples mass, inertia and CoM together; it carries every system-level claim (Tables V, VI, and III). Separately, a *wrench-emulated* step—a virtual force injection that changes the effective translational mass instantaneously, with no inertia or CoM change—is used only where a controlled ablation demands it: isolating the event trigger’s estimation-layer benefit requires a mass discontinuity that is clean and repeatable, free of the grasp-depth scatter and inertia mismatch that a physical attach necessarily introduces (Table II and Fig. 3 only). The supporting weight-schedule ablation runs on the physical gripper, as do all baseline and wind results. The two mechanisms are complementary, not interchangeable, and we keep their scopes disjoint: any conclusion that needs genuine rigid-body coupling is drawn only from the physical experiments (see the red-line note in §VI-A). The wrench mechanism’s inability to change rotational inertia is a documented simulator limitation, not a modeling choice; we state it and its consequences in §VII.

Task state machine: approach (position control) $\rightarrow$ NMPC hand-off and warmup hover $\rightarrow$ controlled descent $\rightarrow$ active grasp enable $\rightarrow$ lift $\rightarrow$ figure-eight $\rightarrow$ phase-triggered release. We state the NMPC scope explicitly: it governs the grasp hover and the post-event adaptation phase; the approach controller is not part of the claim. Headless batch infrastructure with a GCS-heartbeat stand-in; the physical-effectiveness timestamp $t = 0$ is defined by T_{phys} crossing the baseline, and in-band is defined as $\pm 0.08\text{kg}$.

Frozen release configuration. In the evaluated implementation, `c_xy_est_enable` gates the update routine that contains both the legacy mass-domain test and the unified call to `release_decision()`. The node and generic launcher retain `false` as a backward-compatible default, but all 13 release-evaluation batch drivers explicitly set it to `true`. The reported detector results therefore apply to those frozen configurations, not to an unparameterized node launch; without the override, no release decision is evaluated and the controller reaches the conservative UNRESOLVED timeout.

Real-time budget. Both optimizers are acados [24] SQP-RTI (one real-time-iteration SQP step per call, HPIPM [25] QP solver). Table I reports measured wall-clock solve times against the period each loop must meet. The NMPC measurements apply to the deployed 13-state controller: it consumes $\sim 3\%$ of its period at the median and stays under 5% at the 99th percentile. The MHE timing column is retained as a historical measurement of the mass-only 14-state estimator used

TABLE I

MEASURED SOLVE TIMES AND LOOP RATES (SINGLE DESKTOP CORE, SHARED WITH GAZEBO/PX4 SITL). NMPC STATISTICS POOL $n = 1009$ SAMPLES ACROSS THE DEPLOYED 20 Hz BATCHES. MHE STATISTICS ARE FROM AN INSTRUMENTED HISTORICAL MASS-ONLY RUN ($n = 63$ REPORTING WINDOWS OF 20 SOLVES EACH), NOT FROM THE FINAL FIRST-MOMENT ESTIMATOR.

	NMPC	MHE	setpoint
horizon / window	$N = 20, \Delta t = 0.05$ s (1.0 s preview)	$N = 20, \Delta t = 0.1$ s (2.0 s window)	—
state / control dim.	13 / 4	14 / — ^b	—
rate	20 Hz	10 Hz	501
period budget	50 ms	100 ms	201
solve, median	1.3 ms	1.4 ms	—
solve, mean	1.35 ms	1.48 ms	—
solve, p95	2.1 ms	—	—
solve, p99	2.2 ms	—	—
solve, max	15.0 ms ^a	5.1 ms	—
median load	2.6 %	1.4 %	—
p99 load	4.4 %	—	—

^aSingle worst-case sample across all batches (first solve after solver warm-start); the p99 of 2.2 ms is the operative figure. ^bHistorical mass-only MHE; the deployed first-moment MHE has 16 states and requires a separate timing measurement.

TABLE II

EVENT-TRIGGER VS. FIXED WEIGHTS (WRENCH PURE MASS STEP, $n = 6$).

δm	metric	fixed	signal	no-signal
−0.3	settling [s]	1.20	0.30	0.32
	pos_err peak [m]	0.200	0.187	0.173
−0.8	settling [s]	1.57	0.56	0.57
	pos_err peak [m]	0.545	0.483	0.482

by the July/August result batches; it *does not* characterize the final 16-state first-moment estimator. We therefore measured the deployed 16-state estimator separately over the September release batches ($n = 3740$ solves, discarding the first 20 frames of each flight to exclude warm-up): median 9.2 ms, 90th percentile 24.6 ms, 99th percentile 59.2 ms, maximum 72.8 ms against a 100 ms period. The first-moment augmentation is thus roughly $7\times$ slower than the mass-only estimator at the median, and its tail consumes about 60 % of the period. This still meets the deadline in simulation, but it is a desktop core shared with the physics engine, and the margin is narrow enough that we make *no* onboard-headroom claim for the 16-state estimator; establishing one requires measurement on the target computer.

VI. RESULTS

A. Event-trigger benefit (estimation and control layers)

Table II (wrench-emulated pure mass step, $n = 6$). The event trigger cuts settling $\sim 4\times$ ($\delta m = -0.3$ kg) / $\sim 2.8\times$ (-0.8 kg); the **no-signal self-trigger recovers the benefit within noise** (~ 0.02 s of the signal version). Control-layer transient peak drops 6 % to 14 %.

Red line. This experiment reports settling and pos_err peak only; overshoot depth is a wrench inertia-mismatch artifact and

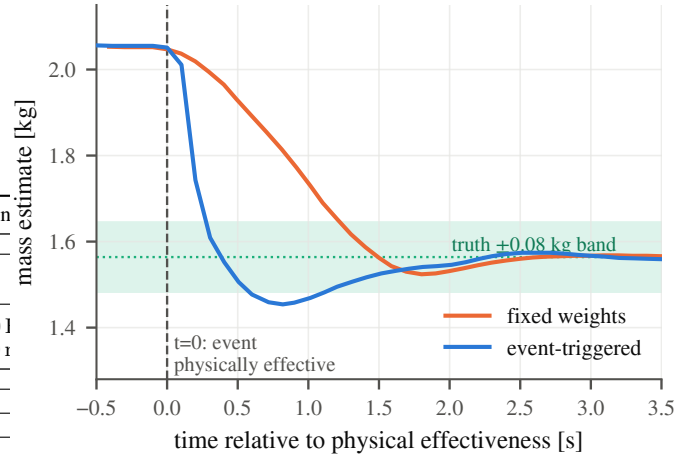


Fig. 3. Mass-estimate timeline (wrench step, $\delta m = -0.5$ kg). The event-triggered estimate reaches the truth band far earlier than fixed weights, at the cost of a transient overshoot below the band before settling—the speed/overshoot trade-off reported honestly.

is not reported here (overshoot conclusions come only from the physical-rigid-body slung experiments).

B. Supporting schedule-tuning ablation

We additionally tested whether the handcrafted transient schedule required numerical optimization. A simulation-in-the-loop cross-entropy search [26] embedded M0 in a five-parameter family and evaluated 96 closed-loop episodes. The surviving candidate was then compared with M0 in a pre-registered paired 2×2 factorial ($n = 8$ per cell). Its median settling-time gain was -0.05 s for a 0.2 kg payload and 0.00 s for a 0.3 kg payload (both non-significant); the test had 97 % power against the larger previously observed effect. We therefore retain M0. This ablation supports a deliberately narrow conclusion: for the tested family, *when* to discard stale stages matters more than the precise weight magnitude. It is not a learning contribution of the deployed system.

C. Eventless grasp-release lifecycle

The controller issues the gripper-release command itself, but deliberately does not use that command to clear its payload model: a commanded release is not proof of physical detachment. We compare an event-driven operational reference (arm A), which resets payload geometry and inner-loop scaling with the command, against the proposed interface (arm B), which consumes only healthy, fresh $m/s/J$ frames. Arm B completes a release only after its no-payload confidence remains above 0.90 for 1.0 s. The two arms were paired under three payload and trajectory conditions. A pre-registered pair was voided and re-flown if either arm crashed before release or lacked payload evidence; these exclusions are reported below and are not absorbed into the completion result.

Among valid pairs, arm B completed autonomous release confirmation in 21/21 flights with no observed post-release instability (Table III). The two-sided 95 % Clopper–Pearson

TABLE III

EVENTLESS RELEASE COMPLETION AND CONFIRMATION DELAY (ARM B, VALID PAIRS ONLY). DELAY RUNS FROM THE RELEASE COMMAND TO THE FRAME AT WHICH THE NO-PAYLOAD CONFIDENCE HAS HELD ABOVE THRESHOLD FOR THE FULL 1.0 s HOLD, SO IT INCLUDES THAT HOLD BY CONSTRUCTION. THE TWO DELAY COLUMNS ARE THE TWO CONTROL PATHS DESCRIBED IN THE TEXT AND ARE *not* TO BE POOLED INTO A SINGLE MEDIAN.

condition	n	compl.	delay median [IQR]	
			in-maneuver	via brake-to-hover
W3: 0.15 kg, 4 m/s	6	6/6	2.78 s [2.76, 2.80]	16.53 s [16.46, 16.60]
W4: 0.30 kg, 4 m/s	7	7/7	7.35 s [7.20, 7.50]	16.40 s [15.76, 16.60]
W5: 0.15 kg, 2 m/s	8	8/8	3.76 s [3.16, 4.46]	16.45 s [16.40, 16.56]
pooled	21	21/21	3.76 s [2.86, 7.20]	16.46 s [16.36, 16.60]

Pooled completion 100 %, two-sided 95 % Clopper–Pearson CI [83.9 %, 100 %]; overall delay max 16.60 s. The share of flights confirming only after the brake-to-hover maneuver is 67 % (W3), 71 % (W4), 50 % (W5) and 62 % pooled; the complement confirms in maneuver.

interval is [83.9 %, 100 %], so this result demonstrates closed-loop feasibility rather than a zero failure rate.

The confirmation delay is *bimodal*, and the two modes are two distinct control paths rather than a spread within one path. In 38 % of flights the no-payload confidence crosses threshold while the vehicle is still flying the figure-eight and confirmation completes in 3.76 s (median). In the remaining 62 % the first mass moment $\hat{s}$ has drifted under sustained excitation, the moment vote is therefore withheld, and the system reaches its 12 s timeout, declares the payload state UNRESOLVED, and executes a 3 s smooth brake to hover; deceleration alone restores observability and confirmation then completes at 16.46 s (median). Reporting one median across both paths would be misleading, so Table III separates them.

In 17 of the 21 flights the payload-presence machine made exactly two transitions—one load declaration at grasp and one empty declaration at release—with no spurious carrying-phase transition. In the four remaining flights, all of which confirmed in-maneuver, the no-payload confidence crossed threshold before the estimator’s presence machine had declared EMPTY, so only the load transition was logged; the release itself was still driven by the estimator interface. Three of the 21 load declarations were raised by the grasp event rather than by sustained mass evidence. The published first moment decayed continuously rather than stepping to zero, confirming that recovery used the intended estimator interface rather than the controller’s release command.

a) Safety boundary: The 21/21 batch validates the interface once its release evidence is accepted; it does not validate a universally robust detector. We therefore tested three physically distinct evidence sources: payload quantity, directional thrust residual, and collapse of the normalized first moment. A fast branch combining quantity and residual without moment evidence declared release prematurely in 3/9 randomized flights during a 4 m/s figure-eight. Loaded-flight mass estimates could remain below the nominal empty threshold for up to 48 s, while a dynamic thrust excursion reached -1.69 N, exceeding the -1.47 N weight change of

TABLE IV
TARGETED RELEASE-SAFETY ABLATION.

decision rule	flights	premature	confirmed / unresolved
quantity + residual fast path	9	3	—
fresh moment-gated paths	12	0	11 / 1

the smallest payload. Neither source is therefore reliable on its own in this regime.

We rejected that branch and require every release decision to include fresh, successful moment-collapse evidence. Ambiguous cases enter UNRESOLVED; a timeout does not clear the model or reset adaptive state. In a subsequent 12-flight smoke test, no flight declared release before the command: 11 releases were confirmed with a median detector delay of 1.9 s, and one remained safely unresolved (Table IV). This is the supported safety conclusion; broader detector robustness remains outside the present claim.

The paired performance comparison is conditional on successful attachment and survival to the release command. Of 77 flights, 20 were voided and re-flown: 17 crashed before release and three lacked MHE payload evidence. The available logs cannot distinguish physical capture failure from estimator non-recognition in those three cases. Among valid pairs, the tail-window tracking RMS is no longer comparable between the arms under the current code: in the 62 % of arm B flights that reach UNRESOLVED, the final 20 s are flown at hover while arm A continues the figure-eight, which accounts for the entire apparent difference. We therefore draw no steady-tracking comparison from this batch. The measurable cost of refusing the event shortcut is the confirmation delay: a median 3.76 s when the moment vote clears in maneuver, and 16.46 s when it does not and the vehicle must first brake to hover.

D. Baselines and ablations

L1-NMPC three-way matrix (Table V). {truth, online, 11} $\times$ {0.2, 0.3} kg $\times$ {0.05, 0.10} m, $n = 5$, 60/60 runs stable, zero divergence. (i) Transient peaks are statistically indistinguishable across the three methods (0.77 m to 1.01 m, within-cell differences 0.01 m to 0.09 m)—L1 is a competent strong baseline and our method is not worse on the transient. (ii) **L1 recovers 15 % to 35 % slower**; the mechanism is that its compensation low-pass cutoff ω_c is bounded by closed-loop stability (0.5 stable, 1.0 unstable on this platform), a bandwidth wall that explicit estimation does not face. (iii) L1 produces no interpretable m/c_{xy} (cannot update thrust-margin constraints), has no estimation-layer settling notion, and still requires the same class prior for gain scaling. Implementation (footnote): L1 is a translational 3-DoF gradient-law augmentation using T_{phys} as the observed thrust; using the commanded thrust self-excites, and a ground low-thrust gate is required.

Wind false-trigger ablation (Table VI). Self-trigger false-positive rate vs. sustained wind magnitude (link-wrench injection, no mass event, $n = 3$). (i) Sharp step at the 1.5 N threshold: sub-threshold vertical wind never false-triggers, supra-threshold always does—a sustained vertical force is

TABLE V
THREE-WAY METHOD $\times$ SCENARIO MATRIX ($n = 5$): POS_ERR PEAK [M] / RECOVERY [S].

cell (mass/ecc)	truth	online	ll
0.2 / 0.05	0.806 / 4.28	0.768 / 3.70	0.821 / 4.26
0.2 / 0.10	0.794 / 3.62	0.810 / 3.92	0.812 / 4.24
0.3 / 0.05	0.834 / 3.56	0.837 / 3.72	0.883 / 4.90
0.3 / 0.10	1.011 / 3.64	0.876 / 3.88	0.891 / 4.54

TABLE VI
WIND FALSE-POSITIVE RATE VS. MAGNITUDE ($n = 3$, NO MASS EVENT).

force [N]	0.5	1.0	1.5	2.0	2.5	3.0
vertical FPR	0/3	0/3	3/3	3/3	3/3	3/3
horizontal FPR	—	0/3	—	0/3	—	0/3

inherently indistinguishable from a mass change on the thrust-residual channel (an honest limitation). (ii) **Immune to horizontal wind**: even at $2\times$ threshold (3 N) horizontal wind never false-triggers, because the detector lives on the collective-thrust channel. (iii) Mitigations: the persistence gate rejects gust pulses; real wind is predominantly horizontal/gusty rather than sustained-DC vertical; the threshold is tunable to the noise floor.

VII. LIMITATIONS AND FUTURE WORK

All system-level evidence is presently SITL-only. Hardware migration replaces simulated motor speeds with ESC telemetry and must re-identify thrust/torque coefficients, noise floors, timing, and gripper compliance before free flight. The vertical arm r_z is unobservable from the thrust-moment channel and remains a weak rack prior; the s_x /pitch channel is noisier than s_y /roll. The deployed diagonal inertia approximation also drops the small $O(r_{xy}^2)$ and yaw-inertia terms.

Release-detector boundary. First-moment augmentation is now part of the default path. The 21/21 paired result shows that the first-moment output can drive complete NMPC recovery without a drop event, but it does not establish distribution-wide detector robustness. The targeted safety ablation identifies the operative boundary: payload quantity and thrust residual can become simultaneously false-empty during dynamic flight, causing premature model clearing in 3/9 flights when combined without moment evidence. The accepted logic therefore requires fresh, healthy moment-collapse evidence and treats timeouts as UNRESOLVED, never as proof of an empty vehicle. After this correction, 12 flights produced no premature declarations, 11 confirmations, and one safe unresolved outcome.

The remaining detector limitation is its moment reference. A running maximum can absorb dynamic estimation excursions above the physically feasible envelope and thereby make an ordinary loaded moment appear collapsed. Future validation will form this reference only from a healthy, low-dynamic, in-envelope interval and freeze it before aggressive flight. Until that revision is validated across broader conditions, our claim

is an eventless, fail-safe control interface rather than a fully robust autonomous release detector.

The current software also has a configuration-coupling defect: the historically named `c_xy_est_enable` parameter is the effective master switch for all release evidence. Although every reported release batch explicitly enabled it, the bare default is `false` and therefore falls through to UNRESOLVED. Before deployment, release decision execution must receive its own explicit enable parameter (or be unconditional in the eventless profile), with startup validation that rejects inconsistent configurations.

Excitation-induced drift of the first moment. The dominant limitation of the release detector is not threshold placement but an observability boundary. Under sustained figure-eight excitation the estimated first moment $\hat{s}$ drifts upward: in affected flights the loaded $\|\hat{s}\|$ grows from 0.012 kg m to 0.031 kg m ($2.5\times$), and the frames that exceed the physical envelope recur with a 12.0 s period, matching the figure-eight half period $\pi/w = 11.1$ s, i.e. the spacing of the lateral-acceleration extrema. While drifted, the moment vote is correctly *withheld* rather than mis-cast, so the failure mode is a delayed confirmation, not a false release. The drift is reversible: deceleration alone restores $\hat{s}$, which collapses below the release threshold within 3.3 s to 3.4 s of leaving the maneuver. Consequently 62 % of flights ($n = 21$) confirm only after the 12 s timeout and the 3 s brake-to-hover, giving a confirmation delay of 16.5 s instead of a few seconds.

We can detect the condition but cannot yet predict it. The count of envelope-exceeding frames (`ovr`), already computed online, separates the two populations sharply: flights with `ovr=0` confirmed in maneuver in 9/9 cases versus 1/7 for `ovr>0` (Fisher $p = 8.7 \times 10^{-4}$). What remains unexplained is *which* flights drift. We pre-registered the hypothesis that drift is driven by maneuver intensity and tested it against the 2 m/s condition, where the affected share should have dropped sharply; it did not (50 % at 2 m/s versus 67 % and 71 % at 4 m/s). We therefore report the intensity explanation as *not supported* and leave the trigger open. A gate on `ovr` would let the vehicle leave the maneuver as soon as the moment channel is known to be unusable, cutting the delay from 16.5 s to approximately 3.5 s; this is implemented in neither the estimator nor the controller and is not claimed here.

Selection and pre-release failures. The main performance tables condition on successful attachment and survival to the release command. Twenty of 77 flights were voided; arm A had 11/39 pre-release crashes and arm B had 6/38, while arm B alone had 3/38 failures of the pre-registered MHE payload-evidence check. The historical logging recorded an attach request but not the simulator joint-state acknowledgement, so those three cases cannot be separated into physical capture failures versus estimator non-recognition. Arm A crashes concentrate in W4 (six of eleven), which exhausted the pre-registered attempt cap at seven valid pairs there rather than the intended eight; W3 stopped at six pairs for scheduling reasons rather than by the cap. Consequently the paper makes no claim that the interface improves grasp, lift, or early figure-eight

stability, and W3/W4 carry somewhat less power than W5. Attachment observability and the pre-release attitude instability require separate experiments rather than reinterpretation of the conditional drop result.

A sustained vertical up/downdraft remains intrinsically confounded with a payload change on a collective-thrust-only residual; the existing wind ablation characterizes this limitation but does not remove it.

Simulator fidelity. We record two known deviations. (i) gz-sim Harmonic provides no supported way to change a link's inertial properties at runtime such that the DART integrator adopts them: `SetComponentData<Inertial>` updates the ECM component only, so introspection tools report the new value while the physics continues to integrate the load-time mass (gz-sim issue #2733). Our wrench mechanism therefore emulates a *translational* mass step by force injection and leaves the rotational inertia unchanged; accordingly it supports only mass-channel claims, and every claim involving inertia, CoM, or eccentricity comes from the physically attached DetachableJoint platform, where the change is genuine (§V). (ii) The eccentric payload is modeled as rigidly attached; a real magnetic gripper admits small compliance and, for a suspended load, pendulum modes this model does not reproduce. Neither deviation affects the sensing-minimal claim, which concerns which signals the estimator consumes rather than how the disturbance is generated.

On learning. Our simulation-in-the-loop optimization of the weight schedule produced no measurable gain (§VI-B), and we identify the binding constraint as the sample economics of high-fidelity evaluation— ~ 85 s per closed-loop episode with no injectable random seed, hence no common random numbers, hence per-candidate noise on the order of the effect being optimized. Approaches that do not share this constraint remain open: supervised targets extracted per-frame from flight data (rather than one scalar loss per episode), or a lower-fidelity differentiable environment for search with SITL reserved for validation. We regard the negative result as delimiting *where* learning can pay off in this pipeline, not as evidence against learned estimator tuning in general.

VIII. CONCLUSION

We presented an eventless payload-adaptive NMPC architecture that uses motor speeds and odometry to update mass, first mass moment, and inertia without consuming attach/drop notifications or treating a release command as proof of physical detachment. Self-triggered MHE window deweighting accelerates abrupt-change recovery, while the atomic, health-gated $m/s/J$ interface and continuous body-rate scheduling connect the estimate safely to the controller. Conditional on successful attachment and pre-release survival, the interface completed release in 21/21 paired SITL flights with no observed post-release instability. It also matched the transient tracking of an L1-NMPC baseline while recovering 15 % to 35 % faster and retaining interpretable physical parameters.

The release-safety ablation defines the claim boundary. Payload quantity and thrust residual became simultaneously

false-empty during aggressive flight, so the accepted design requires fresh moment-collapse evidence and preserves ambiguous cases as UNRESOLVED. This policy produced zero premature declarations in the corrected 12-flight smoke, with 11 confirmed releases and one safe unresolved outcome. The present evidence therefore demonstrates that the payload-event signal can be removed from the adaptive control interface; it does not establish a fully robust release detector. A frozen, robust moment reference, broader SITL validation, and subsequent HITL and flight testing are the next steps.

REFERENCES

- [1] C. V. Rao, J. B. Rawlings, and D. Q. Mayne, "Constrained state estimation for nonlinear discrete-time systems: stability and moving horizon approximations," *IEEE Transactions on Automatic Control*, vol. 48, no. 2, pp. 246–258, 2003.
- [2] L. Meier, D. Honegger, and M. Pollefeys, "PX4: A node-based multithreaded open source robotics framework for deeply embedded platforms," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2015, pp. 6235–6240.
- [3] N. Hovakimyan and C. Cao, $\mathcal{L}_1$ Adaptive Control Theory: Guaranteed Robustness with Fast Adaptation. SIAM, 2010.
- [4] T. Lee, M. Leok, and N. H. McClamroch, "Geometric tracking control of a quadrotor UAV on SE(3)," in *IEEE Conference on Decision and Control (CDC)*, 2010, pp. 5420–5425.
- [5] P. Foehn, A. Romero, and D. Scaramuzza, "Time-optimal planning for quadrotor waypoint flight," *Science Robotics*, vol. 6, no. 56, p. eabh1221, 2021.
- [6] A. Ollero, M. Tognon, A. Suarez, D. Lee, and A. Franchi, "Past, present, and future of aerial robotic manipulators," *IEEE Transactions on Robotics*, vol. 38, no. 1, pp. 626–645, 2022.
- [7] J. Thomas, G. Loianno, J. Polin, K. Sreenath, and V. Kumar, "Toward autonomous avian-inspired grasping for micro aerial vehicles," *Bioinspiration & Biomimetics*, vol. 9, no. 2, p. 025010, 2014.
- [8] D. K. D. Villa, A. S. Brandão, and M. Sarcinelli-Filho, "A survey on load transportation using multirotor UAVs," *Journal of Intelligent & Robotic Systems*, vol. 98, no. 2, pp. 267–296, 2020.
- [9] D. Hanover, P. Foehn, S. Sun, E. Kaufmann, and D. Scaramuzza, "Performance, precision, and payloads: Adaptive nonlinear MPC for quadrotors," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 690–697, 2022.
- [10] E. J. J. Smeur, Q. Chu, and G. C. H. E. de Croon, "Adaptive incremental nonlinear dynamic inversion for attitude control of micro air vehicles," *Journal of Guidance, Control, and Dynamics*, vol. 39, no. 3, pp. 450–461, 2016.
- [11] G. Torrente, E. Kaufmann, P. Foehn, and D. Scaramuzza, "Data-driven MPC for quadrotors," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 3769–3776, 2021.
- [12] K. Sreenath, N. Michael, and V. Kumar, "Trajectory generation and control of a quadrotor with a cable-suspended load — a differentially-flat hybrid system," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2013, pp. 4888–4895.
- [13] T. Tomić, C. Ott, and S. Haddadin, "External wrench estimation, collision detection, and reflex reaction for flying robots," *IEEE Transactions on Robotics*, vol. 33, no. 6, pp. 1467–1482, 2017.
- [14] J. Svacha, J. Paulos, G. Loianno, and V. Kumar, "IMU-based inertia estimation for a quadrotor using Newton–Euler dynamics," *IEEE Robotics and Automation Letters*, vol. 5, no. 3, pp. 3861–3867, 2020.
- [15] V. Wüest, V. Kumar, and G. Loianno, "Online estimation of geometric and inertia parameters for multirotor aerial vehicles," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2019, pp. 1884–1890.
- [16] C. Böhm, M. Scheiber, and S. Weiss, "Filter-based online system-parameter estimation for multicopter UAVs," in *Robotics: Science and Systems (RSS)*, 2021, paper 087, DOI 10.15607/RSS.2021.XVII.087.
- [17] B. Wang, Z. Ma, S. Lai, and L. Zhao, "Neural moving horizon estimation for robust flight control," *IEEE Transactions on Robotics*, vol. 40, pp. 639–659, 2024, open source: RCL-NUS/NeuroMHE.
- [18] W. P. M. H. Heemels, K. H. Johansson, and P. Tabuada, "An introduction to event-triggered and self-triggered control," in *IEEE Conference on Decision and Control (CDC)*, 2012, pp. 3270–3285.

- [19] M. Basseville and I. V. Nikiforov, *Detection of Abrupt Changes: Theory and Application*. Prentice-Hall, 1993.
- [20] L. Hewing, K. P. Wabersich, M. Menner, and M. N. Zeilinger, "Learning-based model predictive control: Toward safe learning in control," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 3, pp. 269–296, 2020.
- [21] L. Brunke, M. Greeff, A. W. Hall, Z. Yuan, S. Zhou, J. Panerati, and A. P. Schoellig, "Safe learning in robotics: From learning-based control to safe reinforcement learning," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 5, pp. 411–444, 2022.
- [22] M. Bangura and R. Mahony, "Nonlinear dynamic modeling for high performance control of a quadrotor," in *Australasian Conference on Robotics and Automation (ACRA)*, 2012.
- [23] N. Koenig and A. Howard, "Design and use paradigms for Gazebo, an open-source multi-robot simulator," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, vol. 3, 2004, pp. 2149–2154.
- [24] R. Verschueren, G. Frison, D. Kouzoupis, J. Frey, N. van Duijkeren, A. Zanelli, B. Novoselnik, T. Albin, R. Quirynen, and M. Diehl, "acados—a modular open-source framework for fast embedded optimal control," *Mathematical Programming Computation*, vol. 14, no. 1, pp. 147–183, 2022.
- [25] G. Frison and M. Diehl, "HPIPM: a high-performance quadratic programming framework for model predictive control," in *IFAC-PapersOnLine (21st IFAC World Congress)*, vol. 53, no. 2, 2020, pp. 6563–6569.
- [26] P.-T. de Boer, D. P. Kroese, S. Mannor, and R. Y. Rubinstein, "A tutorial on the cross-entropy method," *Annals of Operations Research*, vol. 134, no. 1, pp. 19–67, 2005.